14.7.26

Warren Lec 13

Driven damped osc.



$$m\frac{d^2x}{dt^2} = F = \underbrace{-kx}_{F_{spring}} \underbrace{-b\frac{dx}{dt}}_{F_{damping}} + \underbrace{F_0\cos\omega t}_{F_{driving}}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

= natural freq (ang)

$$\ddot{x} = -\frac{k}{m}x - \frac{b}{m}\dot{x} + \frac{F_0}{m}\cos\omega t$$

$$\frac{k}{m} = \omega_0^2$$

γ = damping factor $= \frac{b}{m} = \frac{1}{\tau}$, $f_0 = \frac{F_0}{m}$

(1) $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = f_0\cos\omega t$

$x = x(t)$

steady-state motion of damped and driven osc.

$$x(t) = A\cos(\omega t - \delta)$$

δ: phase lag

Imagine another osc. in y-direction

(2) $\ddot{y} + \gamma\dot{y} + \omega_0^2 y = f_0\sin\omega t$ — driven by

(1) + i(2) →

$$\ddot{z} + \gamma\dot{z} + \omega_0^2 z = f_0 e^{i\omega t}$$

steady-state: $z = \tilde{A}\, e^{i\omega t}$, $(\tilde{A} = Ae^{-i\delta})$

14-17-26

$$\omega_d = \sqrt{\omega_0^2 - \left(\frac{\gamma}{2}\right)^2}$$

= freq. of underdamped osc.

$z = x + iy$

$x = r\cos\theta, \quad y = r\sin\theta$

$r = \sqrt{x^2+y^2} = |z|$

$z = r(\cos\theta + i\sin\theta)$

$= r e^{i\theta}$

$\theta = -\delta$

$\tilde{A} = A_R + iA_I$

$= \underbrace{|\tilde{A}|}_{A} e^{-i\delta} = A e^{-i\delta}$

$z = A e^{i(\omega t - \delta)}$

$\dot{z} = (i\omega) A e^{i(\omega t-\delta)}$

$\ddot{z} = (i\omega)^2 A e^{i(\omega t-\delta)}$

$= -\omega^2 A e^{i(\omega t-\delta)}$

$\left[-\omega^2 A + i\gamma\omega A + \omega_0^2 A\right] e^{i(\omega t-\delta)} = f_0 e^{i\omega t}$

$\left[(\omega_0^2-\omega^2)A + i\underbrace{A\gamma\omega}_{\text{Im part}}\right] = f_0 e^{i\delta} = f_0(\cos\alpha + i\sin\delta)$

$(\omega_0^2-\omega^2)A = f_0\cos\delta$

$\gamma\omega A = f_0\sin\delta$

$\left[(\omega_0^2-\omega^2)^2 + (\gamma\omega)^2\right] A^2 = f_0^2$

$$A = \frac{f_0}{\left[(\omega_0^2-\omega^2)^2 + (\gamma\omega)^2\right]^{1/2}}$$

$$\frac{\gamma\omega}{\omega_0^2-\omega^2} = \tan\delta$$

$$\delta = \tan^{-1}\left(\frac{\gamma\omega}{\omega_0^2-\omega^2}\right)$$

$$A(0) = \frac{f_0}{\omega_0^2}, \quad A(\omega_0) = \frac{f_0}{\gamma\omega_0}$$

$$\frac{A(\omega_0)}{A(0)} = \frac{f_0/\gamma\omega_0}{f_0/\omega_0^2} = \frac{\omega_0}{\gamma} = Q$$

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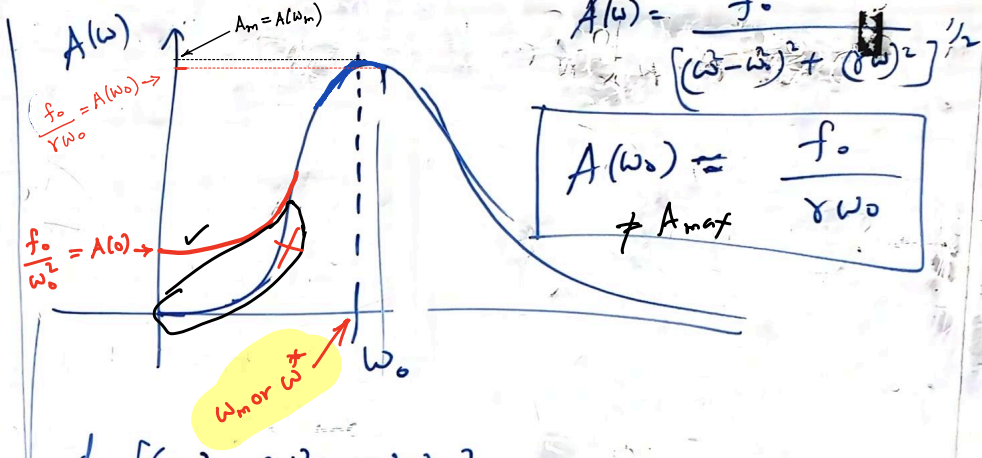
$$\omega_0 = \sqrt{\frac{k}{m}}$$

= natural freq (ang)

$$\ddot{x} = -\frac{k}{m}x - \frac{b}{m}\dot{x} + \frac{F_0}{m}\cos\omega t$$

$$\frac{k}{m} = \omega_0^2$$

γ = damping factor $= \frac{b}{m} = \frac{1}{\tau}$, $f_0 = \frac{F_0}{m}$



$$A(\omega) = \frac{f_0}{\left[(\omega^2-\omega_0^2)^2 + (\gamma\omega)^2\right]^{1/2}}$$

$$A(\omega_0) \simeq \frac{f_0}{\gamma\omega_0} \neq A_{max}$$

$$\frac{d}{d\omega}\left[(\omega^2-\omega_0^2)^2 + \gamma^2\omega^2\right]$$

$$= 2(\omega^2-\omega_0^2)2\omega + \gamma^2 \cdot 2\omega$$

$$= 4\omega\left[\omega^2 - (\omega_0^2 - \frac{\gamma^2}{2})\right] = 0$$

$$\omega = \left(\omega_0^2 - \frac{\gamma^2}{2}\right)^{1/2}$$

$$\omega_m \simeq \omega^* = \left(\omega_0^2 - \frac{\gamma^2}{2}\right)^{1/2} = \omega_0\left[1 - \frac{1}{2}\frac{\gamma^2}{\omega_0^2}\right]^{1/2}$$

For $\gamma \ll \omega_0$ (weak damping)

$$\omega_m \text{ or } \omega^* \simeq \omega_0 \Rightarrow A_{max} \simeq A(\omega_0) = \frac{f_0}{\gamma\omega_0}$$

14/7/26

$A'(\omega) = -\frac{1}{2} \dfrac{2f_0\left[(\omega^2-\omega_0^2)2\omega + \gamma^2\omega\right]}{[\ldots\ldots]^{3/2}}$

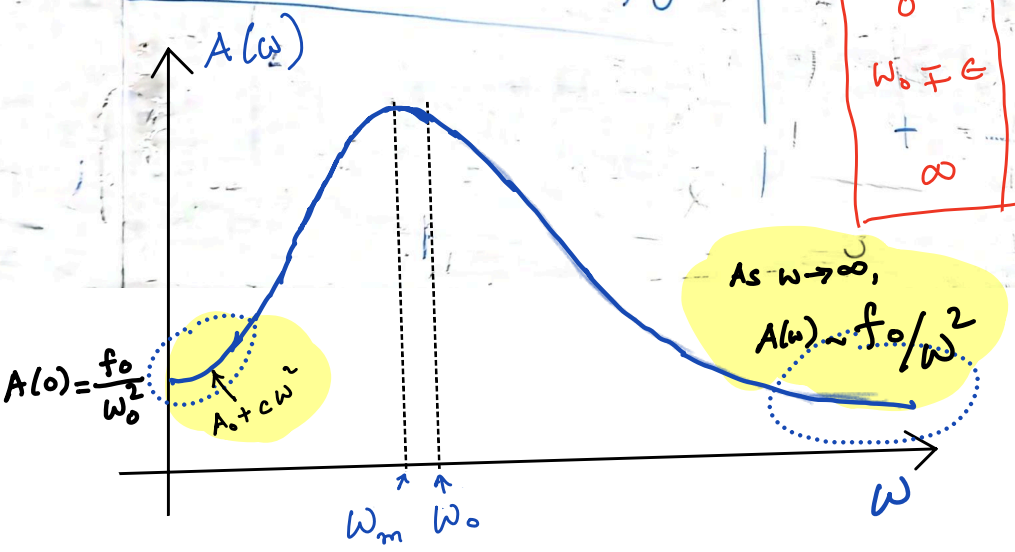
$A'(\omega) = 0$ for max.

$2\omega\left[\omega^2 - \omega_0^2 + \frac{\gamma^2}{2}\right] = 0$

$\omega = 0, \quad \omega = \pm\sqrt{\omega_0^2 - \gamma^2/2}$

$\omega^* = \sqrt{\omega_0^2 - \gamma^2/2}$

$= \omega_0\left[1 - \frac{1}{2}\left(\frac{\gamma}{\omega_0}\right)^2\right]^{1/2}$



$\delta(\omega) = \tan^{-1}\left(\dfrac{\gamma\omega}{\omega_0^2 - \omega^2}\right)$

$0 \le \delta \le \pi$



ω	$\tan\delta$	δ
0	0	0
$\omega_0 \mp \epsilon$	$\pm\infty$	$\mp\frac{\pi}{2}$
$+\infty$	0^-	π

approaches 0 for −ve side

For $\omega = \omega_0 \pm \epsilon$, it can be shown

$\delta(\omega_0 \pm \epsilon) \xrightarrow[\epsilon \to 0]{} \frac{\pi}{2}$

i.e, $\delta(\omega)$ has well defined limit $\frac{\pi}{2}$ at ω_0 and is continuous

$\left[(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2\right] A^2 = f_0^2$

$A = \dfrac{f_0}{\left[(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2\right]^{1/2}}$

$\dfrac{\gamma\omega}{\omega_0^2 - \omega^2} = \tan\delta$

$\delta = \tan^{-1}\left(\dfrac{\gamma\omega}{\omega_0^2 - \omega^2}\right)$

